

Recombination Rate Coefficients in EXHALE: H I, He I, He II, and the He I 2^3S Triplet, with Benchmarks and the Status of Newer Data

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1. Overview

This memo documents how EXHALE treats recombination onto H I, He I, He II, and the metastable He I 2^3S level (the lower level of the He 10830 line), benchmarks the adopted coefficients against other published fits, and reviews whether more recent data warrant an update. The companion to this is the photoionization cross-section memo (photoion_cross_sections.tex); together they cover the two microphysical inputs that set the H/He ionization balance and the He 2^3S population.

The two headline conclusions are: (i) EXHALE's H/He *case-B* coefficients — the Badnell (2023) + Mao & Kaastra (2016) construction adopted as the default on 2026-07-17, with the earlier Hui & Gnedin (1997) fits retained under Legacy_HHe_rates — agree with the Draine (2011) case-B fits to $\sim 2\%$ over 5×10^3 – 2×10^4 K — they are not outdated; the larger differences seen against some escape codes are the *case-A vs. case-B* modeling choice, not data vintage. (ii) EXHALE's He 2^3S triplet recombination (Benjamin, Skillman & Smits 1999, as parametrized by Oklopčić & Hirata 2018) is *identical* to that adopted by the most recent multi-species escape model (Taylor et al. 2025), so it reflects current escape-community practice.

2. What EXHALE uses

All H/He recombination coefficients are defined in src/modules/radiation/Cool_coeff.f90 and assembled in ionization_equilibrium.f90. Units are $\text{cm}^3 \text{s}^{-1}$.

Default case-B coefficients (Badnell 2023 + Mao & Kaastra 2016). Since 2026-07-17 the default H/He case-B coefficients are built as

$$\alpha^B(T) = \alpha_{\text{Badnell}}^A(T) - \alpha_1^{\text{Mao}}(T), \quad (1)$$

the Badnell radiative total minus the Mao & Kaastra ground-state ($n = 1$) coefficient, so that direct recombination to the ground state — the case-B (“on-the-spot”) exclusion, whose re-emitted photon is assumed re-absorbed locally — is removed. The Badnell radiative fit (2023 update of Badnell 2006) is

$$\alpha^{\text{RR}}(T) = \frac{A}{\sqrt{T/T_0} (1 + \sqrt{T/T_0})^{1-b} (1 + \sqrt{T/T_1})^{1+b}}, \quad b = B + C e^{-T_2/T}, \quad (2)$$

and the Mao & Kaastra (2016, A&A 587, A84) $n = 1$ coefficient is

$$\alpha_1(T) = a_0 \times 10^{-10} \frac{\vartheta^{-b_0 - c_0 \ln \vartheta} (1 + a_2 \vartheta^{-b_2})}{1 + a_1 \vartheta^{-b_1}}, \quad \vartheta = k_B T / \text{eV}, \quad (3)$$

with the coefficient sets given in `Cool_coeff.f90` (`rr_badnell`, `rr_mao`). For He II → He I the Badnell three-term dielectronic recombination (`dr_HeII_badnell`; `adf48`, negligible below $\sim 5 \times 10^4$ K) is added to α^A before the subtraction. He III → He II is a hydrogenic ion, but it is *not* built by rescaling the H II → H I coefficient: `alphaB_HeIII_new` evaluates its own Badnell RR parameter set and subtracts its own Mao & Kaastra $n = 1$ term. Only the legacy branch below uses the explicit $Z = 2$ rescaling. These are the same coefficients MoCHII uses (`src/recomb_mod.f90` of the separate MoCHII code — not a file of this repository); see `docs/atomic_data_EXHALE_vs_MoCHII`.

Legacy option (Hui & Gnedin 1997, case B). Setting the input flag `Legacy_HHe_rates`: `True` reverts the H/He case-B coefficients to the Hui & Gnedin (1997) fits inherited from ATES. With $\lambda_i = 2T_i/T$ and T_i the ionization temperature (157807, 285335, 631515 K for H I, He I, He II),

$$\alpha_{\text{HII}}^B(T) = 2.753 \times 10^{-14} \frac{\lambda_{\text{H}}^{1.5}}{[1 + (\lambda_{\text{H}}/2.74)^{0.407}]^{2.242}}, \quad (4)$$

$$\alpha_{\text{HeIII}}^B(T) = 2 \times 2.753 \times 10^{-14} \frac{\lambda_{\text{He}^+}^{1.5}}{[1 + (\lambda_{\text{He}^+}/2.74)^{0.407}]^{2.242}}, \quad (5)$$

the second being the hydrogenic scaling of the first, with the He II total $\alpha_{\text{HeII}}^B = 1.26 \times 10^{-14} \lambda_{\text{He}}^{0.75}$. The new default and this legacy branch agree to $\sim 1\%$ for H II and He III; the He II value differs more, because the new coefficient carries the Badnell dielectronic term explicitly (Tables 1 and 2).

He II → He I and the 2³S network. When the He 2³S network is *off*, He II → He I uses the total case-B coefficient above (the Badnell+DR–Mao default, or the Hui & Gnedin $1.26 \times 10^{-14} \lambda_{\text{He}}^{0.75}$ under `Legacy_HHe_rates`). When the network is *on*, the He II → He I recombination is split into singlet and triplet channels (`ionization_equilibrium.f90`; the total case-B value is overwritten), following Benjamin, Skillman & Smits (1999) as parametrized by Oklopčić & Hirata (2018):

$$\alpha_1(T) = 1.54 \times 10^{-13} (T/10^4)^{-0.486} \quad (\text{singlet, } 1^1S), \quad (6)$$

$$\alpha_3(T) = 2.10 \times 10^{-13} (T/10^4)^{-0.778} \quad (\text{triplet, } 2^3S). \quad (7)$$

These are *effective* (cascade-summed) coefficients: α_3 is the total rate at which recombinations populate the metastable 2³S level via direct capture plus radiative cascades through the higher triplet terms. The triplet captures the majority of He recombinations, $\alpha_3/(\alpha_1 + \alpha_3) \simeq 0.53\text{--}0.63$ over $5 \times 10^3\text{--}2 \times 10^4$ K (Table 2).

One further overwrite follows in the same routine and is on by default. With `He_rec_coupling: True` (the default), `he_rec_coupling` replaces the singlet coefficient again, by $y\alpha_1 + 0.25\alpha_B$ when the triplet network is on and by $\alpha_B + y\alpha_1$ when it is off, where y is the fraction of the ground-capture continuum that ionizes H rather than He and the $0.25\alpha_B$ term is the singlet-excited capture channel the 2³S network does not carry. So the number the ionization balance sees for He II→He I is that combination, not α_1 alone; α_1 and α_3 above are the underlying channel data. Setting `He_rec_coupling: False` leaves the split of the previous paragraph in place.

Metals are not revisited here. Most of them use the Badnell radiative + dielectronic recombination (`alpha_rec_metal`); the two exceptions in the code are Ca I, a Shull & Van Steenberg (1982) power law, and the two iron stages, which take the Huang et al. (2023) Eqs. (5)–(6) fit because the Badnell dielectronic project does not reach the Mn-like and Cr-like sequences. See `docs/atomic_data_EXHALE_vs_MoCHII`.

3. H⁺ → H: case-A vs. case-B, not data vintage

Figure 1 and Table 1 compare EXHALE’s case-B α_{HII} with the Draine (2011) case-A/B fits and with the total (case-A) coefficient adopted by the recent escape model of Taylor et al. (2025, their Storey & Hummer 1995 value). EXHALE tracks the Draine *case-B* curve to $\sim 2\%$ across the whole range. The Taylor+2025 value is $\sim 40\text{--}90\%$ higher, but that is because it is a *case-A* (total) rate: codes that transport the diffuse recombination continuum explicitly use case A, whereas EXHALE’s on-the-spot scheme correctly uses case B. The underlying data agree; the difference is the radiative-transfer treatment.

Table 1: H⁺ → H recombination coefficient ($10^{-13}\text{cm}^3\text{s}^{-1}$). The “EXHALE (B)” column is the legacy (`Legacy_HHe_rates`) Hui & Gnedin fit; the Badnell–Mao default now in force gives 4.531, 2.606, 1.856 and 1.446 on the same four rows.

T [K]	EXHALE (B)	Draine (B)	Draine (A)	Taylor+25 (A)
5000	4.543	4.428	6.733	6.608
10000	2.592	2.540	4.130	4.241
15000	1.837	1.818	3.087	3.271
20000	1.428	1.428	2.505	2.721

4. Helium recombination

The left panel of Fig. 2 repeats the case-A/B story for He⁺⁺ → He⁺: EXHALE (case B) matches the hydrogenic Draine case-B to $\sim 2\%$, while the Verner & Ferland (1996) total used by Taylor+2025 is $\sim 45\%$ higher (case A). The right panel shows He⁺ → He: the Hui & Gnedin total case-B (the legacy `Legacy_HHe_rates` value; the Badnell–Mao default

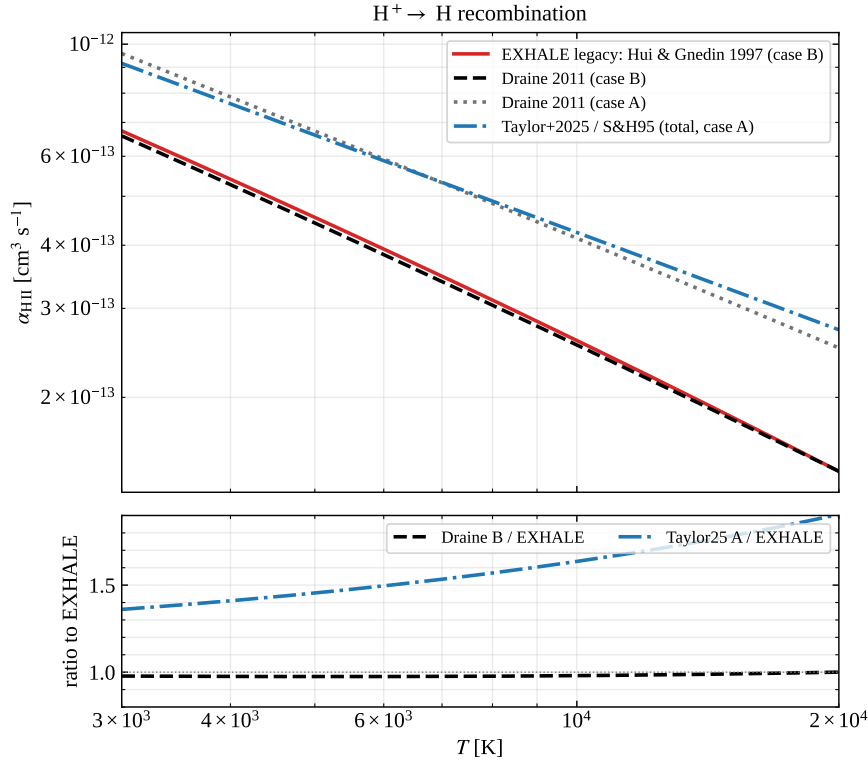


Figure 1: $H^+ \rightarrow H$ recombination. The red curve is the legacy Hui & Gnedin (1997) case-B fit (Legacy_HHe_rates); the current default (Badnell–Mao case B) differs by only $\sim 1\%$ and would overlie it. It overlaps the Draine (2011) case-B fit to $\sim 2\%$ (bottom panel); the case-A (total) curves of Draine and Taylor+2025 are $\sim 40\text{--}90\%$ higher. The offset is the case-A vs. case-B choice, not a data update.

is $\sim 7\%$ higher here, Table 2), and the Benjamin+1999 singlet/triplet split that EXHALE uses when the 2³S network is active. Table 2 lists the values.

Two points are worth noting. (i) The triplet recombination α_3 (the quantity that feeds He 10830) is *identical* in EXHALE and in the latest escape models — both adopt Benjamin+1999. (ii) The Benjamin+1999 effective total $\alpha_1 + \alpha_3$ is $\sim 40\%$ larger than the Hui & Gnedin (1997) total case-B (e.g. 3.64 vs. $2.62 \times 10^{-13} \text{ cm}^3 \text{ s}^{-1}$ at 10^4 K); the two come from different methods (effective level-resolved cascade sums vs. a recombination-spectrum case-B total). When the 2³S network is on, EXHALE consistently uses the Benjamin split for both channels, so the He ionization balance and the metastable source term are computed from the same data set.

5. Is there newer data?

H I and He II (hydrogenic). Essentially solved. Hui & Gnedin (1997), Storey & Hummer (1995), Verner & Ferland (1996), and Badnell (2006/2023) agree to $\sim 1\text{--}2\%$ (Fig. 1). On 2026-07-17 EXHALE moved its default onto the Badnell (2023) radiative total minus the Mao & Kaastra (2016) $n = 1$ coefficient (case B), matching MoCHII; the Hui & Gnedin fits remain available through Legacy_HHe_rates. The shift is $\sim 1\%$ for H II and He III and larger for He II, which now

Table 2: He recombination coefficients ($10^{-13} \text{ cm}^3 \text{ s}^{-1}$). Left: $\text{He}^{++} \rightarrow \text{He}^+$. Right: $\text{He}^+ \rightarrow \text{He}$ — Hui & Gnedin total (case B) and the Benjamin+1999 singlet α_1 , triplet α_3 , and triplet fraction. The “EXHALE(B)” and “HG97 tot” columns are the legacy (Legacy_HHe_rates) Hui & Gnedin fits; the current Badnell–Mao default gives $\alpha_B(\text{He II}) = 2.807$ and $\alpha_B(\text{He III}) = 15.36$ at 10^4 K (see docs/atomic_data_EXHALE_vs_MoCHII).

$T \text{ [K]}$	$\text{He}^{++} \rightarrow \text{He}^+$			$\text{He}^+ \rightarrow \text{He}$			
	EXHALE(B)	Draine(B)	Tayl25(A)	HG97 tot	α_1	α_3	$\alpha_3/(\alpha_1+\alpha_3)$
5000	25.61	25.35	34.51	4.400	2.157	3.601	0.625
10000	15.45	15.13	22.61	2.616	1.540	2.100	0.577
15000	11.37	11.09	17.66	1.930	1.265	1.532	0.548
20000	9.09	8.86	14.82	1.556	1.100	1.225	0.527

carries the Badnell dielectronic term explicitly. The only remaining meaningful choice is case A vs. B (§3.).

He I total and the 2³S triplet — no update available. The newer He I references often cited here — **Porter, Ferland, Storey & Detisch (2012, MNRAS 425, L28; 2013 erratum MNRAS 433, L89)** (the Cloudy default) and **Del Zanna & Storey (2022)** — are *emissivity* papers. They improve the He I *line cascade / line-ratio* predictions, but they do *not* provide, and do not change, the quantity EXHALE needs: the *total* (effective) recombination into the triplet ladder, α_3 , which all funnels to the metastable 2³S. That total is set by (case-B total recombination) $\times$ (triplet spin fraction 3/4) — ultimately by the photoionization cross sections (Milne) and spin statistics — and is *robust* across modern treatments:

- EXHALE (Benjamin+1999): $\alpha_3 = 2.10 \times 10^{-13} T_4^{-0.778}$, $\alpha_1 = 1.54 \times 10^{-13} T_4^{-0.486} \text{ cm}^3 \text{ s}^{-1}$;
- 2025 thermosphere model (arXiv:2509.14499; from Draine 2011 §14.3): $\alpha_3 = \frac{3}{4} (2.72 \times 10^{-13} T_4^{-0.789}) = 2.04 \times 10^{-13} T_4^{-0.789}$, with the *identical* singlet $\alpha_1 = 1.54 \times 10^{-13} T_4^{-0.486} \text{ cm}^3 \text{ s}^{-1}$.

α_3 agrees to $\sim 2\text{--}3\%$ and α_1 is identical. The most recent multi-species escape model (Taylor et al. 2025) likewise adopts Benjamin+1999 for the triplet. Attempting to extract a different value from the local Cloudy install confirmed the point: its `he_iso_recomb.dat` stores the direct radiative recombination for each level, and summing all triplet ($S = 3$) captures *is* the effective $\alpha(2^3S)$ in case B, but the file carries no usable level $\rightarrow (n, \ell, S)$ map for its 1642 He levels (only 111 are resolved in `helike_energies.dat`) and a trial sum overshoots the known case-A total by $\sim 2\times$ — i.e. there is no clean drop-in Cloudy number either.

Recommendation. **Leave the recombination as it is.** EXHALE’s Benjamin+1999 α_1, α_3 already match the latest escape-modeling practice and the underlying atomic physics to $\lesssim 3\%$; the Porter/Del Zanna “updates” are line-emissivity improvements that do not alter the total 2³S recombination. The H/He case-B rates were themselves updated on 2026-07-17 to the Badnell (2023) + Mao & Kaastra (2016) construction (a $\sim 1\%$ shift for H II/He III; the legacy Hui & Gnedin fits remain under Legacy_HHe_rates). The other worthwhile atomic-data update in this round is the

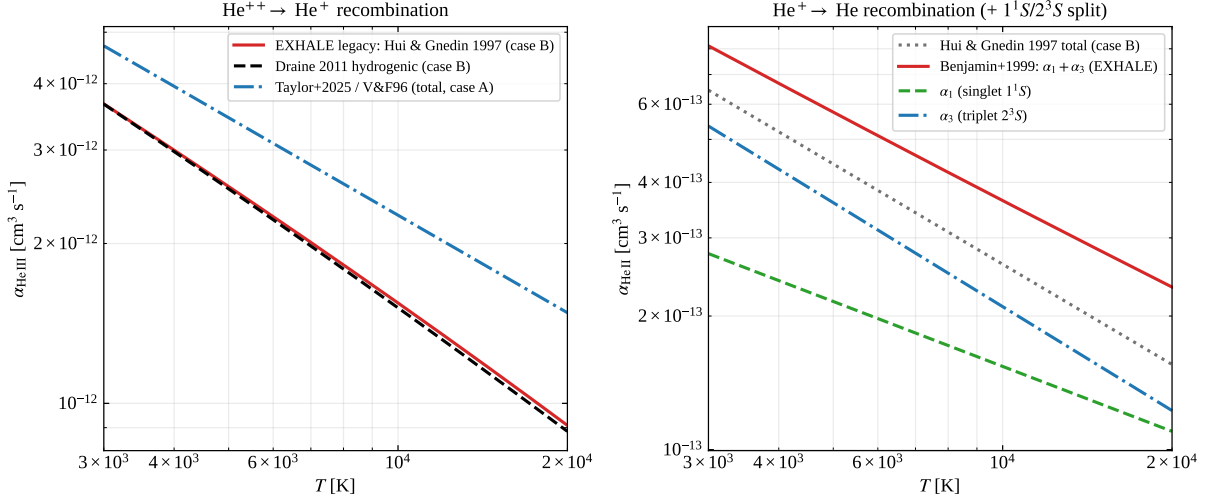


Figure 2: Helium recombination. **Left:** He⁺⁺ → He⁺ — the legacy Hui & Gnedin case B (red, Legacy_HHe_rates) $\approx$ Draine hydrogenic case B (black); Verner & Ferland 1996 / Taylor+2025 total (case A) is $\sim 45\%$ higher. The Badnell–Mao default lies within $\sim 1\%$ of the red curve here. **Right:** He⁺ → He — Hui & Gnedin total case B (dotted; the default is $\sim 7\%$ higher) and the Benjamin+1999 singlet/triplet split ($\alpha_1 + \alpha_3$, α_1 , α_3) used by EXHALE when the 2³S network is active.

photoionization of the He I ground state (Verner+1996, now the EXHALE default; see docs/Update_EXHALE §22), which shifts the He 10830 depth by $\sim 3\%$.

6. Reproducing this comparison

```
cd EXHALE_v1.00/docs
python3 compare_recombination.py
# -> figures/rec_H.pdf, figures/rec_He.pdf
```

The script ports the EXHALE coefficients from Cool_coeff.f90 verbatim and codes the Draine (2011), Verner & Ferland (1996) / Taylor+2025, and Benjamin+1999 reference fits.